

	1	2	3	4	5	6
A	STATUS: DRAFT – UNVERIFIED – requires human review at Gate G2 MicroScout flight controller – root sheet. Sub-sheets: Power, MCU and USB, Sensors and expanders, Camera, LEDs, buzzer, receiver. Generated by tools/sch from hardware/*/design/*.py – edit the Python, not this file, until G3 hand-off. Connectivity: net labels and power symbols; check report in review/G2/.					A
B						B
C						C
D						D
	1	2	3	4	5	6

Power



File: power.kicad_sch

MCU and USB



File: mcu.kicad_sch

Sensors and expanders



File: sensors.kicad_sch

Camera, LEDs, buzzer, receiver



File: io.kicad_sch

Generated schematic – not reviewed. Values are first-pass G2 choices.

STATUS: DRAFT – UNVERIFIED – requires human review at Gate G2

MicroScout (open hardware, CERN-OHL-S-2.0)

Sheet: /

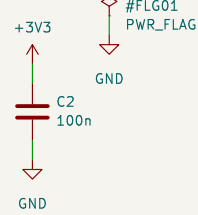
File: microscout-fc.kicad_sch

Title: MicroScout flight controller – root

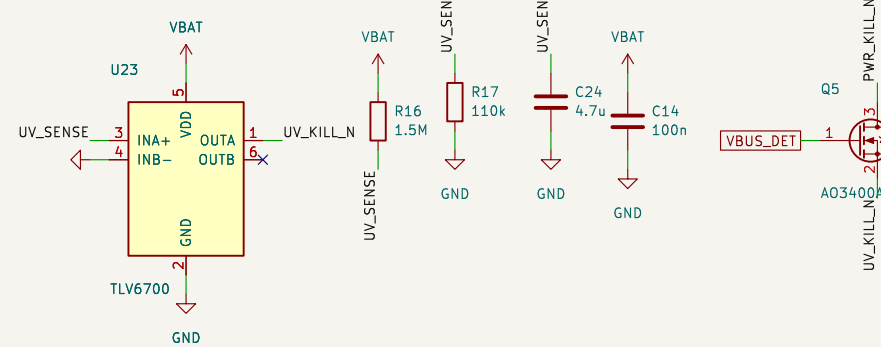
Size: A4	Date: 2026-10-07	Rev: G2-draft-1
KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4	Id: 1/5

MicroScout flight controller – Power

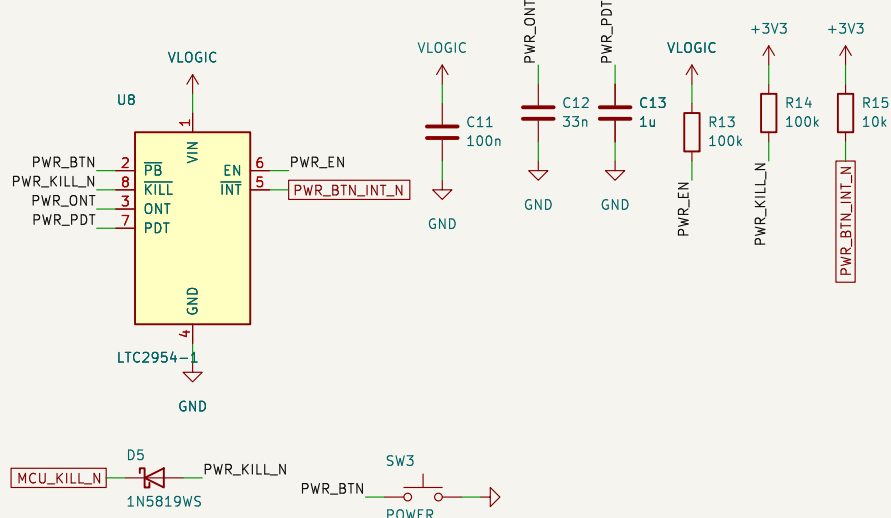
XT30 pin 1 = pack +, pin 2 = pack - (footprint polarity to confirm at G3).
RS1 high-side 1 mOhm: INA226 full scale 81.92 mV / 1 mOhm = 81.9 A; LSB 2.5 mA (budgets 4a)
Input filter per INA226 datasheet 6.4.2: ≤ 10 Ohm per input + 0.1 uF differential.



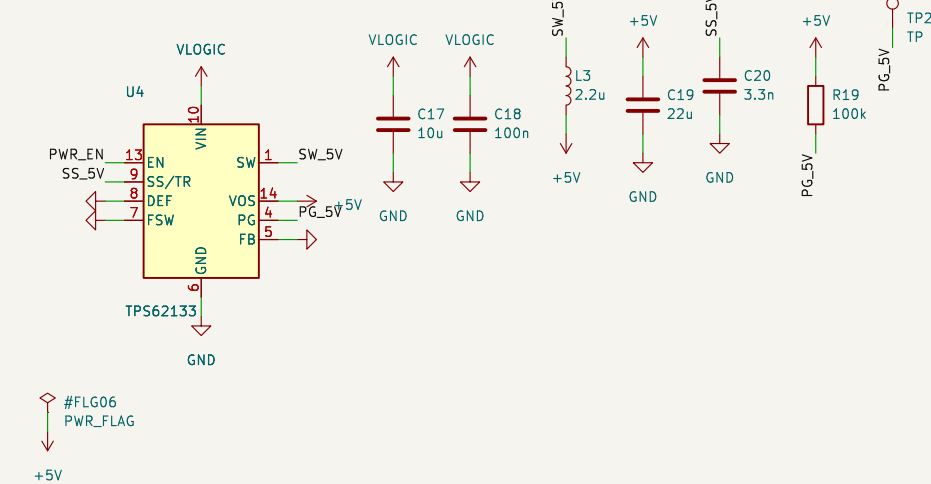
Trip: $V = 0.3945 \text{ V} \times (1.5 \text{ M} + 110 \text{ k}) / 110 \text{ k} = 5.77 \text{ V}$ (2.89 V/cell); release $0.400 \times 14.64 = 5.85 \text{ V}$.
C24 filters load sag: $\tau = (1.5 \text{ M} || 110 \text{ k}) \times 4.7 \text{ uF} = 0.48 \text{ s}$, so throttle punches do not cut power.
Q5 disconnects the cut-off while USB is present (VBUS_DET low); TLV6700 holds OUTA low below its UVL.
VBAT is not 0 V without a battery (charger battery detection, D3 leakage), which would block USB-only p.
Firmware lands first at 3.4 V/cell under load.

[illegible][illegible]

EN (open drain) enables the 3.3 V and 5 V bucks and the ESC gate-driver supply switch U24 (D-036).
KILL: 100 k pull-up to 3V3; pulled low by the MCU through D5 (GPIO48 cannot switch KILL high, D-055) or by the undervoltage detector U23 through Q5 (disables while USB is present, D-055). VF05 (D-33) at 3.3 uA -0.2 V < 0.57 V.
L_ON = 0.033 uF / 1.56e-4 uF/ms + 1 ms = 213 ms press to turn on; C_PDT 1 uF -> 33 uA power-down window.



Loads: ELRS receiver, 4x WS2812B, LED buffer, buzzer. FSW low = 2.5 MHz; DEF low = nominal 5.0 V.

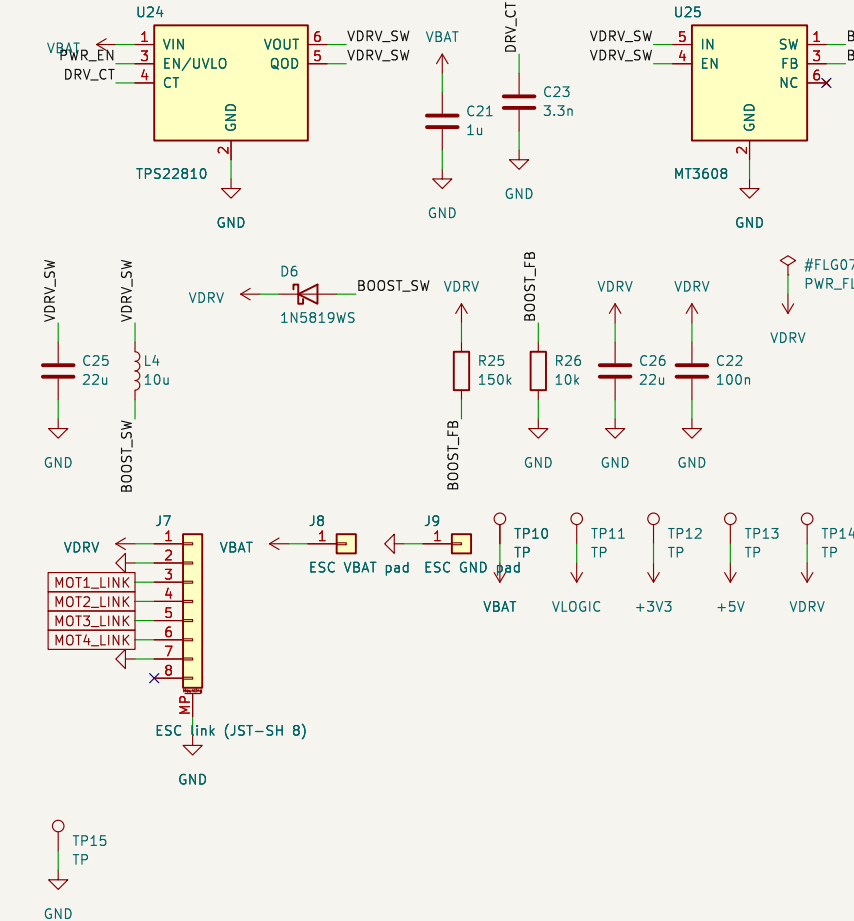


VDRV = 9.6 V for the 4x D6288Q drivers and the ESC board's own 3.3 V regulator (D-048). A boost keeps the drivers well above their 5.0 V UVLO when a 25 pack sags under load, and gives 10 V-class gate drive.

VOUT = $0.6 \text{ V} \times (1 + 150 \text{ k} / 10 \text{ k}) = 9.6 \text{ V}$. $D = 1 - V_{in}/V_{out} = 0.08 - 0.38$. Load -50 mA (ESC 3.3 V + drivers).

Ripple $\delta = 6 \text{ V} \times 0.38 / (10 \mu\text{H} \times 1.2 \text{ MHz}) = 0.19 \text{ A p-p}$, far below the 4 A switch limit.

QOD on VOUT discharges VDRV_SW when off; the boost then passes 0 V. CT 33 nF: -0.7 ms rise.



Sheet: /Power/
File: power.kicad_sch

Size: A2	Date: 2026-10-07	Rev: G2-draft-1
----------	------------------	-----------------

KiCad E.D.A. kicad-cli 7.0.11+dfsg-1

10	11	
----	----	--

STATUS: DRAFT – UNVERIFIED – requires human review at Gate G2

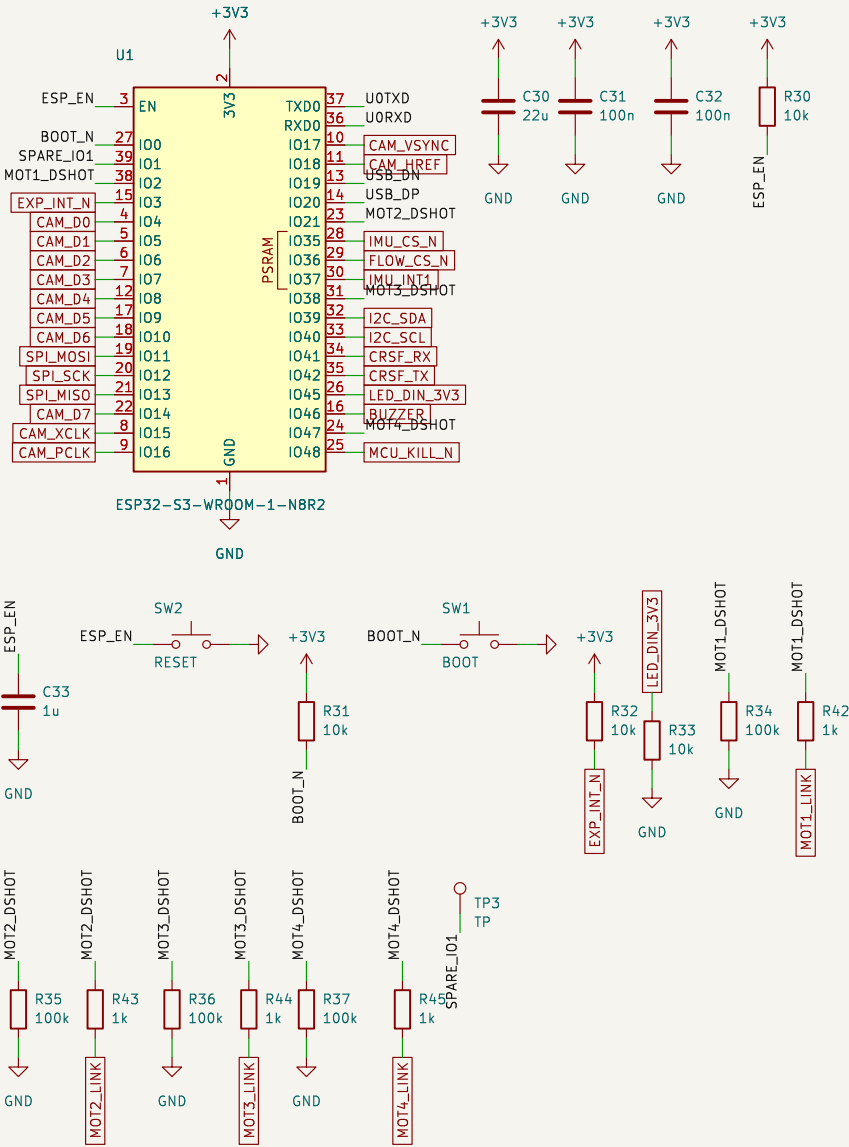
MicroScout flight controller – MCU and USB

ESP32-S3-WROOM-1-N8R2 (pin table: review/G1/pin-allocation.csv; module datasheet Table 3-1, section 9)

3V3: 22 uF + 2x 0.1 uF (datasheet peripheral schematic). EN: 10 k + 1 uF RC (datasheet section 9).

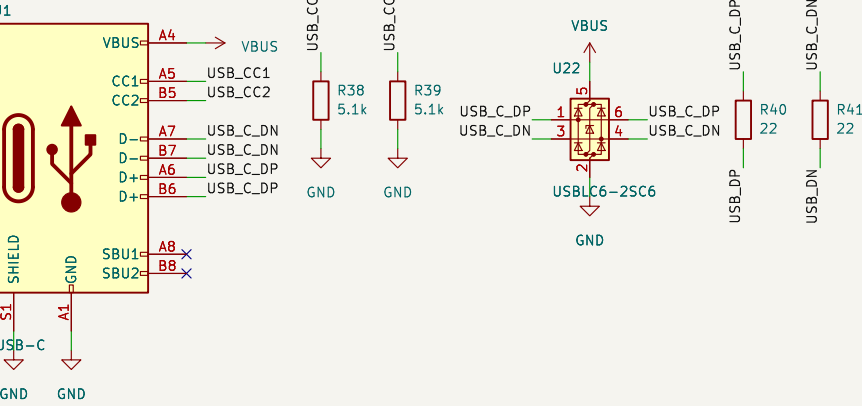
Strapping: GPIO0 10 k up (BOOT button to GND), GPIO3 10 k up, GPIO45 10 k down, GPIO46 100 k down (at Q3).

GPIO1 is spare (test point). DShot outputs have 100 k pull-downs (motor stopped during reset) and 1 k series resistors that limit current into an unpowered ESC MCU (e.g. USB-only power). GPIO48 reaches KILL through D5.



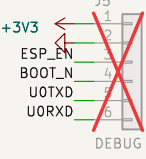
USB-C (device, USB 2.0) with ESD protection

CC1/CC2 5.1 k to GND (sink, Rd). USBLC6-2SC6 at the connector; 22 Ohm series resistors near the module (ESP32-S3 checklist 'initial value 22/33 Ohm'). VBUS feeds the charger and the logic OR-ing only.



Debug pads (UART0 ROM log, EN, IO0) – optional 1.27 mm header, not fitted

Normal flashing and logging use native USB (GPIO19/20).



Generated schematic – not reviewed. Values are first-pass G2 choices.
STATUS: DRAFT – UNVERIFIED – requires human review at Gate G2

MicroScout (open hardware, CERN-OHL-S-2.0)

Sheet: /MCU and USB/
File: mcu.kicad_sch

Title: MicroScout flight controller – MCU and USB

Size: A3
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4

Date: 2026-10-07

Rev: G2-draft-1

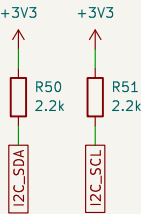
Id: 3/5

STATUS: DRAFT – UNVERIFIED – requires human review at Gate G2

MicroScout flight controller – Sensors and expanders

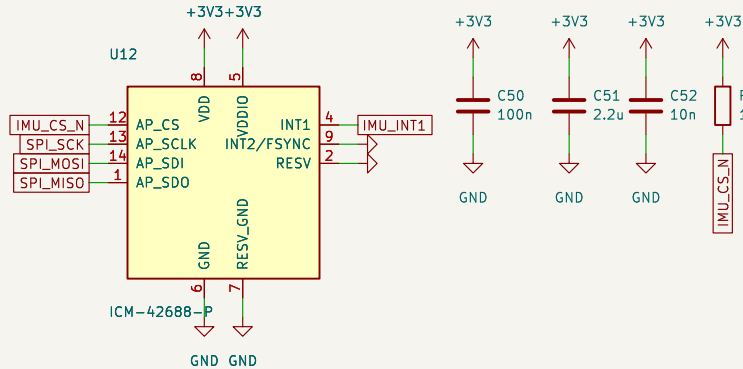
I2C bus (400 kHz) pull-ups

2.2 k: rise time 0.8473 x 2.2 k x C_{bus}; 300 ns limit -> C_{bus} <= 160 pF. Estimate ~150 pF with 4 ToF wires (G2 estimate, measure at G6; 1.8 k is the fallback). Devices: 0x20/0x21 expanders, 0x29 ToF (re-addressed), 0x2C mag, 0x30 camera SCCB, 0x40 INA226, 0x6B charger (datasheet also says 0x6A), 0x76 baro.
The Flow deck's VL53L1x sits at 0x29 with no XSHUT; firmware re-addresses it first, before releasing any satellite.



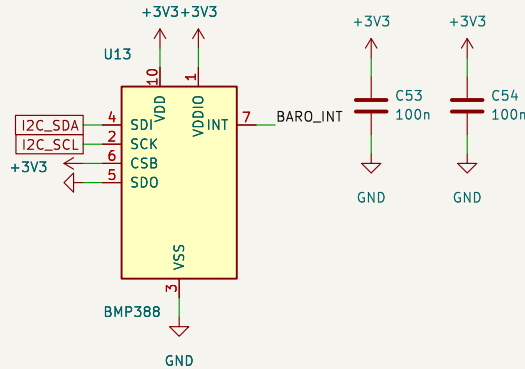
IMU – ICM-42688-P on SPI2 (datasheet DS-000347 Table 10/11, Fig. 7)

VDD 0.1 uF + 2.2 uF, VDDIO 10 nF, RESV pins to GND; INT2/FSYNC to GND (unused), 4-wire SPI.
CS pull-up 10 k keeps the IMU deselected while the ESP32 boots.



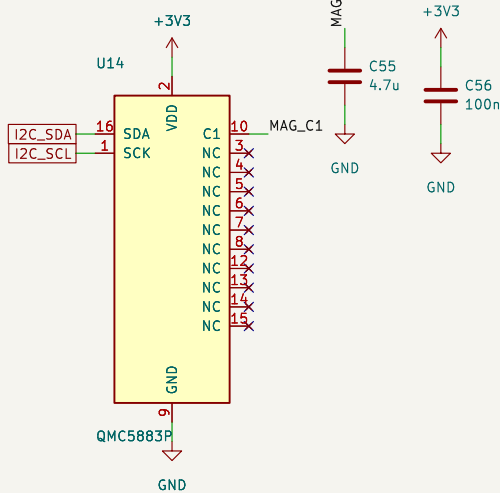
Barometer – BMP388 fitted (BMP390 same footprint; BMP390 out of stock 2026-10-07, OQ-6)

I2C: CSB to VDDIO, SDO to GND -> 0x76. 100 nF on VDD and VDDIO (datasheet Fig. 25).



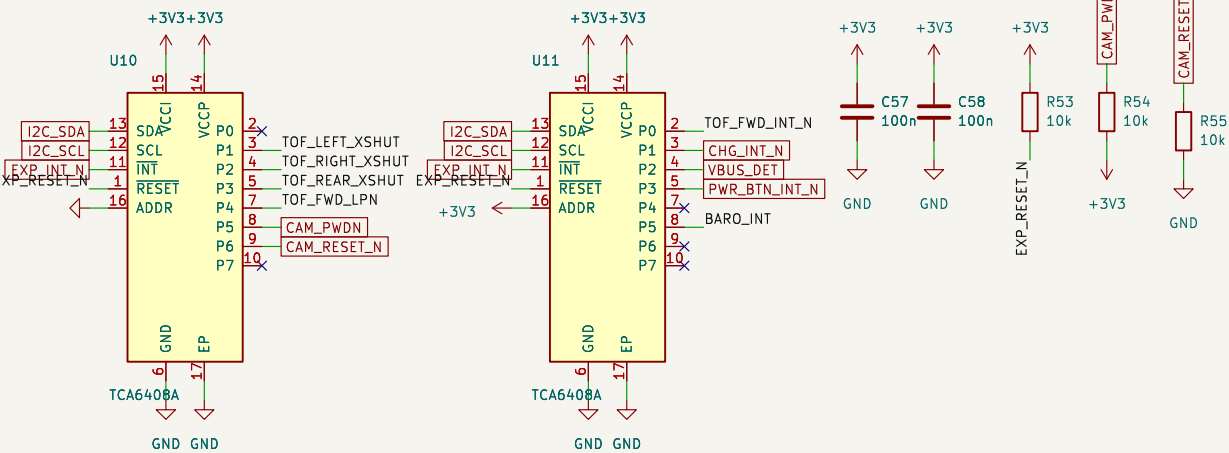
Magnetometer – QMC5883P (address 0x2C fixed)

C1 pin: 4.7 uF, ESR < 200 mOhm (datasheet 4.3.3). 100 nF VDD decoupling added by the agent (not in the datasheet).



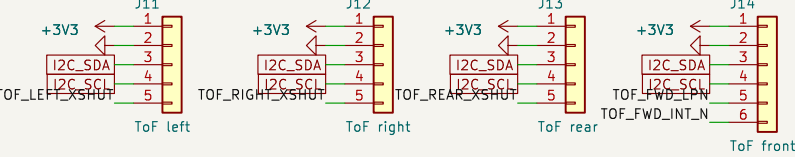
I2C GPIO expanders – TCA6408A x2 (0x20 outputs, 0x21 inputs); RESET pulled up, INT wired-OR to GPIO3

Expander pins power up as inputs, so the external pulls set the safe state: ToF in reset (XSHUT low, LPn low on the satellite), camera powered down (PWDN high) and held in reset.
LPn low on the satellite, camera powered down (PWDN high) and held in reset.



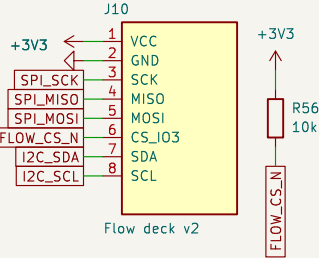
ToF satellite boards (hardware/tof-side x3, hardware/tof-front x1) – wire pads

Side boards: VL53L1X, XSHUT 10 k pull-down on the satellite. Front board: VL53L5CX, LPn/INT/I2C_RST pulls on the satellite. Firmware re-addresses every sensor at boot (all power up at 0x29).



Optical flow – Bitcraze Flow deck v2 (PMW3901 + VL53L1x), wired to pads (OQ-4)

Deck pins used: VCC, GND, SCK, MISO, MOSI, IO_3 (PMW3901 CS), SDA, SCL (Bitcraze deck pinout).
Fed from 3.3 V; the Crazyflie supplies 3.0 V – whether 3.3 V is within the deck's limits is UNCONFIRMED (G2 check).
CS pull-up 10 k keeps the flow sensor deselected during boot.



Generated schematic – not reviewed. Values are first-pass G2 choices.
STATUS: DRAFT – UNVERIFIED – requires human review at Gate G2

MicroScout (open hardware, CERN-OHL-S-2.0)

Sheet: /Sensors and expanders/
File: sensors.kicad_sch

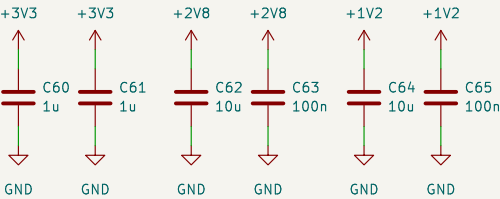
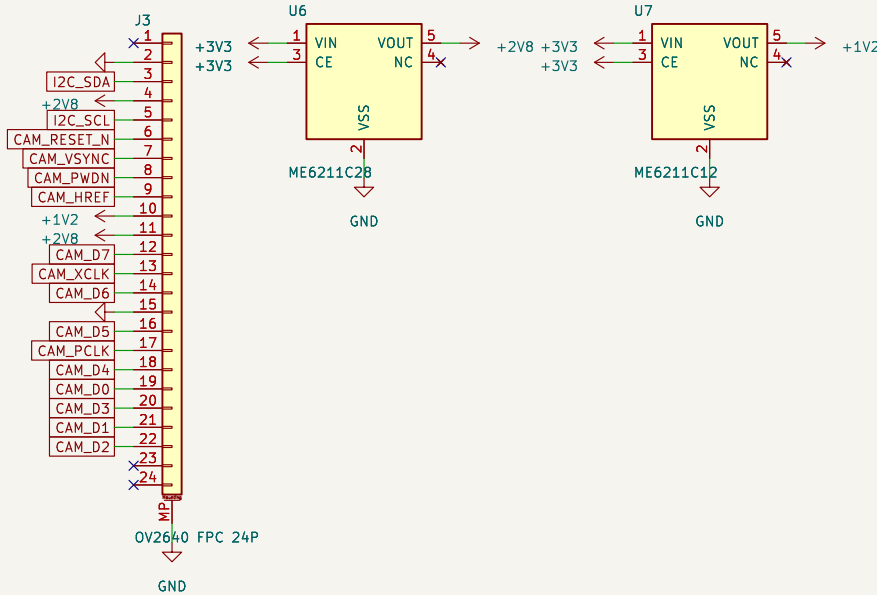
Title: MicroScout flight controller – Sensors and expanders

Size: A3 Date: 2026-10-07 Rev: G2-draft-1
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 Id: 4/5

STATUS: DRAFT – UNVERIFIED – requires human review at Gate G2

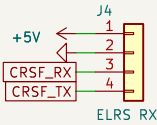
MicroScout flight controller – Camera, LEDs, buzzer, receiver

Camera – OV2640 on a 24-pin 0.5 mm FPC (pinout: Espressif ESP32–S3–EYE MB v2.2, J4)
AVDD + DOVDD 2.8 V (as ESP32–S3–EYE); DVDD 1.2 V (OV2640 datasheet core 1.14–1.26 V; the S3–EYE uses 1.5 V – conflict noted, G2 check against the purchased module). Rails are always on with 3V3 (D–049); PWDN (pulled high) keeps the sensor in standby. SCCB is on the main 3.3 V I2C bus (as on the S3–EYE) – level margin is a G2 check.



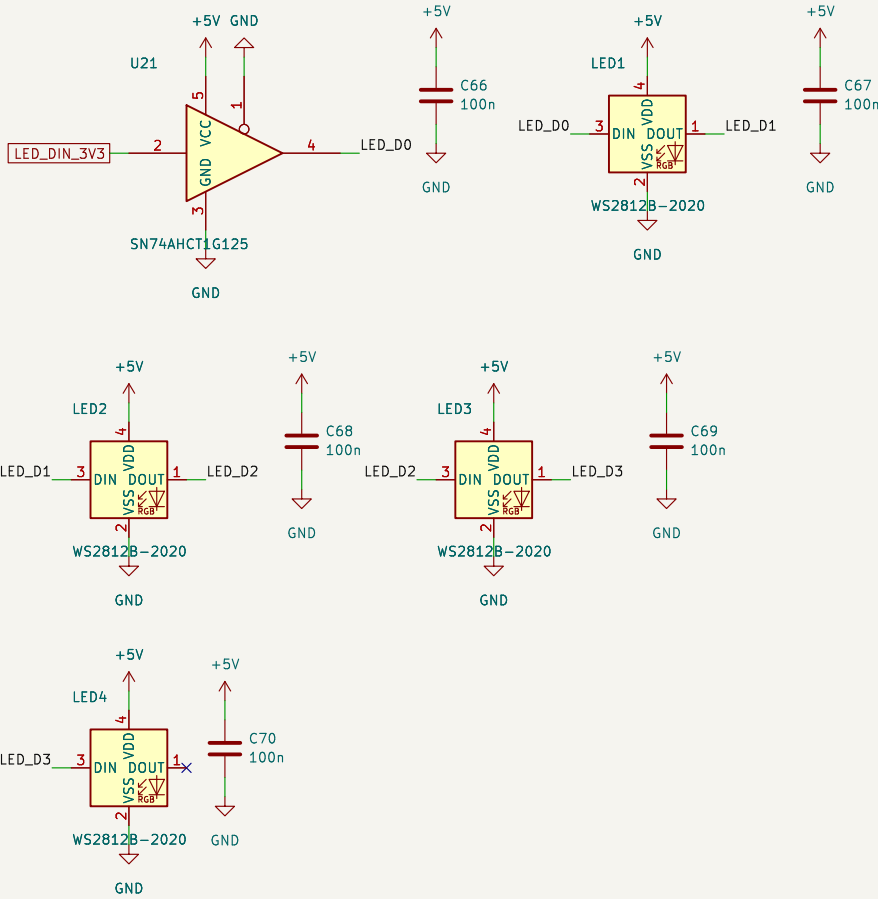
ExpressLRS receiver pads (CRSF, 420 kbaud) – BetaFPV ELRS Lite or similar, powered from 5 V

Receiver TX -> GPIO41 (CRSF_RX); receiver RX <- GPIO42 (CRSF_TX).



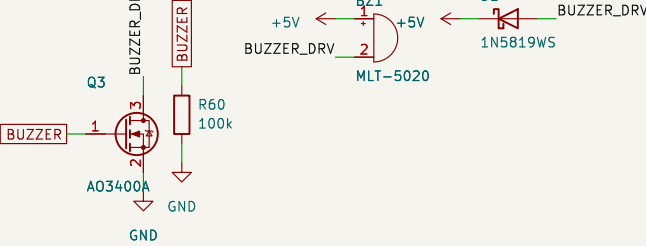
Status LEDs – 4x WS2812B–2020 on 5 V; SN74AHCT1G125 shifts 3.3 V data to 5 V (WS2812B VIH = 0.65 VDD)

Data from GPIO45 via SPI3 MOSI (D–042). 100 nF per LED (agent choice; the LED datasheet says none is needed).



Buzzer – magnetic MLT–5020 driven by AO3400A from 5 V (rated voltage UNCONFIRMED; G2 check)

GPIO46 is a strapping pin that must be low at boot: 100 k gate pull–down keeps it low and the FET off.



Generated schematic – not reviewed. Values are first-pass G2 choices.
STATUS: DRAFT – UNVERIFIED – requires human review at Gate G2

MicroScout (open hardware, CERN–OHL–S–2.0)

Sheet: /Camera, LEDs, buzzer, receiver/
File: io.kicad_sch

Title: MicroScout flight controller – Camera, LEDs, buzzer, receiver

Size: A3 Date: 2026–10–07

Rev: G2–draft–1

KiCad E.D.A. kicad-cli 7.0.11+dfsg–1build4

Id: 5/5